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Efficient method for robot skill learning

Abstract

A method and system for robot skill learning for high precision assembly tasks employing a compliance controller. A reinforcement learning (RL) controller is pre-trained in an offline mode using human demonstration data, where several repetitions of the demonstration are performed while collecting state and action data for each repetition. The demonstration data is used to pre-train a neural network in the RL controller, with no interaction of the RL controller with the compliance controller/robot system. Following pre-training, the RL controller is moved to online production where it is coupled to the compliance controller/robot system in a self-learning mode. During self-learning, the neural network-based RL controller uses action, state and reward data to continue learning correlations between states and effective actions. Co-training is provided as needed during self-learning, where a human operator overrides the RL controller actions to ensure successful assembly operations, which improves the learned performance of the RL controller.

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Background/Summary

BACKGROUND

Field

(1) The present disclosure relates generally to a method for robot skill learning and, more particularly, to a method for robot skill learning applicable to high precision assembly tasks employing a compliance controller, where a reinforcement learning controller is first pre-trained in an offline mode using human demonstration data, and then update self-training is performed online with human override action when necessary to ensure that the self-learning has a sufficient number of successful executions.

Discussion of the Related Art

(2) The use of industrial robots to repeatedly perform a wide range of manufacturing and assembly operations is well known. However, some types of tight-tolerance assembly operations, such as

installing a peg into a hole or plugging one part into another, are still problematic for robots to perform. These types of operation are often performed manually because robots have difficulty detecting and correcting the complex misalignments that may arise in tight-tolerance assembly tasks. That is, because of minor deviations in part poses due to both grasping and fixturing uncertainty, the robot cannot simply move a part to its nominal installed position, but rather must “feel around” for the proper alignment and fit of one piece into the other.

(3) In order to make assembly tasks robust to these inevitable positioning uncertainties, robotic systems typically utilize force controllers (aka compliance control or admittance control) where force and torque feedback is used to provide motions commands needed to complete the assembly operation. A traditional way to set up and tune a force controller for robotic assembly tasks is by manual tuning, where a human operator programs a real robotic system for the assembly task, runs the program, and adjusts force control parameters carefully in a trial and error fashion. However tuning and set up of these force control functions using physical testing is time consuming and expensive, since manual trial and error has to be performed. Also, when tuning is performed on real systems, re-design may be required if the real system does not meet requirements. Parameter tuning on real physical test systems is also hazardous, since robots are not compliant, and unexpected forceful contact between parts may therefore damage the robot, the parts, or surrounding fixtures or structures.

(4) Systems exist for tuning force control parameters for robotic assembly in a simulation environment, but these existing systems exhibit several limitations including slowness and lack of robustness. Other existing systems attempt to apply learning functions to a force controller. However, existing imitation learning systems are not robust, and failure data typically overwhelms the demonstration data in the learning controller. Existing reinforcement learning systems take a long time to train, and typically experience many failed trials along the way, with risk of the part damage and hazardous behavior discussed earlier.

(5) In view of the circumstances described above, improved methods are needed for robotic assembly skill learning in tight tolerance applications using compliance controllers.

SUMMARY

(6) The following disclosure describes a method and system for robot skill learning applicable to high precision assembly tasks employing a force or compliance controller. A reinforcement learning controller is first pre-trained in an offline mode using human demonstration data, where several repetitions of the human demonstration are performed while collecting state and action data for each demonstration repetition. The demonstration data is used to pre-train a neural network in the reinforcement learning controller, with no interaction of the reinforcement learning controller with the compliance controller/robot system during pre-training. Following initial pre-training, the reinforcement learning controller is moved to online production where it is coupled to the compliance controller/robot system in a self-learning mode. During self-learning, the neural network-based reinforcement learning controller uses action, state and reward data to continue learning correlations between states and effective actions. Co-training is provided as needed during self-learning, where a human operator overrides the reinforcement learning controller actions to ensure successful assembly operations, which in turn improves the learned performance of the reinforcement learning controller.

(7) Additional features of the present disclosure will become apparent from the following description and appended claims, taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an illustration of a robotic assembly operation being performed on tight-tolerance

parts, illustrating sources of part positioning uncertainty which create challenges for robotic assembly operations;

(2) FIG. 2 is an illustration of parts being robotically assembled, where the parts require alignment in a manner which causes the robot to perform a hole search in a plane perpendicular to the insertion axis;

(3) FIG. 3 is an illustration of parts being robotically assembled, where the parts require alignment in a manner which causes the robot to perform a phase search to find a proper rotational orientation about the insertion axis;

(4) FIG. 4 is a block diagram illustration of a system configured for a robotic assembly operation using a compliance controller (i.e., force or admittance control), as known in the art;

(5) FIG. 5 is a block diagram illustration of a system configured for robotic assembly skill learning, using a compliance controller with an actor-critic reinforcement learning module, according to an embodiment of the present disclosure;

(6) FIG. 6 is a block diagram illustration of a reinforcement learning system, as used in embodiments of the present disclosure;

(7) FIG. 7 is a block diagram illustration of a system configured for robotic assembly skill learning, using a compliance controller and robot with a trained actor-only reinforcement learning module, according to an embodiment of the present disclosure;

(8) FIG. 8 is a flowchart diagram of a method for robotic assembly skill learning using actor-critic reinforcement learning with a compliance controller/robot system, according to an embodiment of the present disclosure;

(9) FIG. 9 is a conceptual illustration of a system for robot assembly skill learning combining human demonstration and reinforcement learning-based discovery, according to an embodiment of the present disclosure;

(10) FIG. 10 is a block diagram illustration of a system configured for robotic assembly skill learning, using a compliance controller with a reinforcement learning module, in a co-training mode for human correction during ongoing self-learning, according to an embodiment of the present disclosure;

(11) FIG. 11 is a block diagram illustration of a system configured for robotic assembly skill learning, using a compliance controller with a reinforcement learning module, in an online self-learning mode, according to an embodiment of the present disclosure; and

(12) FIG. 12 is a flowchart diagram of a method for robotic assembly skill learning using a reinforcement learning controller with a compliance controller/robot system, including offline pre-training using human demonstration data and online self-learning with human co-training, according to an embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(13) The following discussion of the embodiments of the disclosure directed to a method for robot skill learning using a reinforcement learning controller pre-trained offline using demonstration data and update self-trained online with human co-training is merely exemplary in nature, and is in no way intended to limit the disclosed techniques or their applications or uses.

(14) The use of industrial robots for a wide variety of manufacturing and assembly operations is well known. The present disclosure is directed to overcoming the challenges encountered in many robotic assembly operations.

(15) FIG. 1 is an illustration of a robotic assembly operation being performed on tight-tolerance parts, illustrating several sources of part positioning uncertainty which create challenges for robotic assembly operations. A robot **100** having a gripper **102** grasps a first part **110** which is to be assembled with a second part **120**. In this example, the first part **110** is a peg part, and the second part **120** is a hole structure. The peg part **110** is to be inserted into a hole in the hole structure **120**. The tolerances of the parts in a peg-in-hole assembly are typically quite tight, so that the assembly can operate without excessive looseness after assembled. Some peg-in-hole assemblies have dual

coaxial pegs on one part, or dual parallel-axis pegs on one part, which must be simultaneously inserted into dual holes on the other part, which makes the assembly operation even more difficult. Many other types of mating part assemblies—such as electrical connectors, complex planar shapes, etc.—exhibit similarly tight tolerances.

(16) The types of assembly operations described above are often performed manually because robots have difficulty detecting and correcting the complex misalignments that may arise in tight-tolerance assembly tasks. That is, because of minor deviations in part poses, the robot cannot simply move a part to its nominal installed position, but rather must “feel” the alignment and fit of one piece into the other. There are many possible sources of errors and uncertainty in part poses. First, the exact position and orientation (collectively, “pose”) of the peg part **110** as grasped in the gripper **102** may vary by a small amount from the expected pose. Similarly, the exact pose of the hole part **120** in its fixture may also vary from the expected pose. In systems where a camera **130** is used to provide images of the workspace scene for location identification, perception error can also contribute to the uncertainty of relative part positioning. In addition, calibration errors in placement of the robot **100** and the fixture holding the part **120** in the workspace, and minor robot joint position variations, can all further contribute to part positioning uncertainty. These factors combine to make it impossible for the robot **100** to simply pick up the peg part **110** and insert it in a single motion into the hole structure **120**.

(17) FIG. **2** is an illustration of parts being robotically assembled, where the parts require alignment in a manner which causes the robot to perform a hole search in a plane perpendicular to the insertion axis. A gripper **202** grasps a part **210** which must be inserted into a hole **220** in the same manner as shown in FIG. **1**. A distance **230**, exaggerated for visual effect, represents the uncertainty in the lateral position of the part **210** relative to the hole **220**. In order to find the proper alignment of the part **210** with the hole **220**, the robot may be required to perform a hole search, where the gripper **202** moves the part **210** back and forth in a zig-zag pattern **240** in a plane which is perpendicular to the axis of the part **210**.

(18) FIG. **3** is an illustration of parts being robotically assembled, where the parts require alignment in a manner which causes the robot to perform a phase (rotational) search to find a proper rotational orientation about the insertion axis. A gripper **302** grasps a first part **310** which must be mated with a second part **320**. In this case, the first and second parts **310/320** are gears having teeth which must be precisely aligned. In order to find the proper alignment of the part **310** with respect to the part **320**, the robot may be required to perform a phase search, where the gripper **302** finely adjusts the rotational position of the part **310** about its pivot axis while attempting to lower the part **310** into position and engagement with the part **320**.

(19) Although the hole search and phase search capabilities described above exist in robotic systems, they can be inefficient, and they are often ineffective when the part assembly involves anything more complex than a simple hole alignment or rotational alignment. The illustrations of FIGS. **1-3** are merely exemplary, and many other types of tight-tolerance installation and assembly tasks exist which are difficult to perform robotically.

(20) In order to make assembly tasks robust to these inevitable positional uncertainties, robotic systems typically utilize force control based functions to “feel” for the proper part fit. A traditional way to set up for robotic assembly tasks is by manual tuning, where a human operator programs a real robotic system for the assembly task, runs the program, and adjusts force control parameters in a trial and error fashion. However, tuning and set up of these force control functions using physical testing is time consuming and expensive due to the manual trial and error which has to be performed. Parameter tuning on real physical test systems is also hazardous, since robots are not compliant, and unexpected forceful contact may damage robots, workpieces, and/or surrounding fixtures or structures.

(21) Systems exist for tuning force control parameters for robotic assembly in a simulation environment, but these existing systems exhibit several limitations. Some of these systems are

designed to simulate only one specific type of assembly task—such as one of those shown in FIGS. 1-3—and cannot be applied to general assembly tasks because the techniques include pre-programmed motion strategies which are specific to the particular type of assembly task. Other existing systems for parameter tuning in a simulation environment still require significant human experience and expertise for guiding the selection of force control parameters used in the simulations.

(22) The present disclosure describes methods for robot skill learning using improved reinforcement learning techniques which overcome the drawbacks of existing methods of programming or teaching a robot to perform assembly tasks. One disclosed method uses a reinforcement learning controller pre-trained offline using human demonstration data and update self-trained online with human override action (co-training), and another disclosed method uses an actor-critic reinforcement learning controller which is trained and then used in actor-only mode after training. Both of these disclosed methods are discussed in detail below.

(23) FIG. 4 is a block diagram illustration of a system 400 configured for a robotic assembly operation using a compliance controller (i.e., force or admittance control), as known in the art. In the physical world, a robot controller 410 communicates with a robot such as the robot 100 of FIG. 1. The controller 410 may communicate with the robot 100 via a cable 412 as shown, or wirelessly. The controller 410 provides joint motion commands to the robot 100 and receives state feedback from the robot 100, as known in the art and discussed below. As illustrated in FIG. 1, the robot 100 has the gripper 102 grasping the first part 110, and the controller 410 provides commands with the objective of assembling the first part 110 with (into) the second part 120. The illustrations of the parts 110 and 120 are merely exemplary. Compliance controllers may be used for assembly of any components requiring high precision—such as those of FIGS. 1-3, a dual peg part with a dual hole part, electrical connectors, etc.

(24) A block 420 represents the controller 410 and the robot 100 in block diagram form. The controller 410 is configured as a compliance controller, the functions of which are discussed below. A block 430 provides a nominal target position of the first part 110. The nominal target position could be predefined and unchanging for a particular robot workcell, or the nominal target position could be provided by a vision system based on an observed position of the second part 120, or by some other means. For the sake of this discussion, it is assumed that the position of the second part 120 in the robot workcell is known, and the nominal target position from the block 420 defines the position of the first part 110 to install it into the second part 120. The nominal target position of the first part 110 may of course then be transformed to gripper coordinates, which can then be converted to robot joint positions using inverse kinematics in a known manner.

(25) A summing junction 440 is included after the block 430. Although the junction 440 does not have a second input in FIG. 4, a second input will be added in later figures. A block 450 defines motion limits of the robot 100. The motion limits ensure that the robot 100 does not take an excessively large motion step during the control process, where excessively large motion could create a hazardous situation or result in forceful contact between the robot 100 and/or the parts 110/120 with each other or with other objects in the workcell. If the difference between the target position and the current position is greater than the motion limit, then the motion limit will prevail and limit the size of the step.

(26) A block 460 includes an admittance control function which interacts with the robot in a block 470 performing the assembly task in a block 480. The blocks 470 and 480 represent the physical actions of the robot 100 as it installs the first part 110 into the second part 120. The robot in the block 470 provides state feedback on a line 472 to the admittance control function in the block 460. The state feedback provided on the line 472 includes robot joint states (position/velocity), along with contact forces and torques. Alternately, position and velocity state data may be provided in Cartesian coordinates, which can readily be converted to joint coordinates, or vice versa, via the transformation calculations described above. A force and torque sensor (not shown) is required in

the operation of the robot **100** to measure contact forces between the parts **110** and **120**, where the force and torque sensor could be positioned between the robot **100** and the gripper **102**, between the gripper **102** and the first part **110**, or between the second part **120** and its “ground” (fixing device). The contact force and torque can also be measured from robot joint torque sensors or estimated from other signals such as motor currents.

(27) The admittance control function in the block **460** operates in the following manner, as known in the art and discussed only briefly here. Impedance control (or admittance control) is an approach to dynamic control relating force and position. It is often used in applications where a manipulator interacts with its environment and the force-position relation is of concern. Mechanical impedance is the ratio of force output to motion input. Controlling the impedance of a mechanism means controlling the force of resistance to external motions that are imposed by the environment. Mechanical admittance is the inverse of impedance—it defines the motions that result from a force input. The theory behind the impedance/admittance control method is to treat the environment as an admittance and the manipulator as an impedance.

(28) Using the target position from the junction **440** and the motion limit from the block **450**, the admittance control function in the block **460** computes a target velocity (in six degrees of freedom) to move the workpiece from its current position to the target position (or the motion limited step size). The admittance control function then computes a command velocity by adjusting the target velocity with a force compensation term, using an equation such as: $V = V_{sub.d} + K_{sub.v.sup.-1} F$, where V is the command velocity vector (this equation applies to translational motion), $V_{sub.d}$ is the target velocity vector, $K_{sub.v.sup.-1}$ is the inverse of an admittance gain matrix, and F is the measured contact force vector from the force sensor fitted to the robot or the workpiece. The vectors all include three translational degrees of freedom in this example. A similar equation is used to compute rotational command velocities ω using contact torque feedback.

(29) The command velocities computed as described above are then converted to command joint velocities $\{\dot{q}\}_{sub.cmd}$ for all robot joints by multiplying the inverse of a Jacobian matrix J by the transpose of the command velocities vector, as follows: $\{\dot{q}\}_{sub.cmd} = J_{sup.-1} [V, \omega]_{sup.T}$. A low pass filter may also be provided after the computation of the command joint velocities to ensure smoothness and feasibility of the commanded velocities. The computed command joint velocities are provided to the robot, which moves and measures new contact forces, and the target position is again compared to the current position and the velocity calculations are repeated. Using the force feedback and the robot state feedback on the line **472**, the admittance control function in the block **460** repeatedly provides motion commands to the robot in the block **470** in attempting to reach the target position from the junction **440**.

(30) The elements **430-460** are programmed as an algorithm in the controller **410**. The interaction of the controller **410** with the robot **100** occurs via the cable **412** (or wirelessly) in the real world, and this is represented in the block **420** by the forward arrow (motion commands) from the block **460** to the block **470** and the feedback line **472** (joint states and contact forces).

(31) As discussed earlier, traditional compliance control techniques are not effective for all types of assembly tasks. For example, in tight-tolerance assembly operations, the robot in the block **470** may spend a long time in the feedback loop with the admittance control function in the block **460**, and may never ultimately complete the assembly task, including the possibility of part damage in the process. This situation may be somewhat alleviated by fine tuning of the impedance/admittance control parameters, but this is only effective in some situations, and only for a particular workpiece assembly operation. The techniques of the present disclosure have been developed to address these shortcomings of existing compliance controllers.

(32) FIG. 5 is a block diagram illustration of a system **500** configured for robotic assembly skill learning, using a compliance controller and robot with an actor-critic reinforcement learning module, according to an embodiment of the present disclosure. At the top of FIG. 5 is the block **420** depicting the compliance controller interaction with the robotic assembly operation, as illustrated in

FIG. 4 and discussed above. The depiction of the controller **410** and the robot **100** are omitted from FIG. 5, as the roles of these physical elements corresponding to the compliance control block **420** have already been explained in connection with FIG. 4.

(33) In the technique of the present disclosure, a reinforcement learning module **510** is coupled to the compliance controller/robot system in the block **420**. The purpose of the reinforcement learning module **510** is to learn an effective high level motion strategy for the assembly operation and use this strategy to guide the compliance controller/robot system. The specific technique used in the reinforcement learning module **510** is an actor-critic reinforcement learning structure, including an actor block **520** and a critic block **530**. FIG. 5 depicts the reinforcement learning module **510** as initially configured in training mode. A later figure depicts how the trained reinforcement learning system is deployed in inference mode for ongoing operations of the compliance controller/robot system.

(34) FIG. 6 is a block diagram illustration of a reinforcement learning system **600**, as used in embodiments of the present disclosure. An environment block **610** represents the physical system—in this case, that is a compliance controller/robot performing an assembly task. An agent **620** is the reinforcement learning module. The basic concept of a reinforcement learning system is to observe the operational states of a system, along with the effectiveness of the system, and learn which input actions to the system lead to the best results.

(35) To accomplish this, the agent **620** receives state data $S_{sub.t}$ (t being a time step) on line **630**, and reward data $R_{sub.t}$ on a line **640**. For a robotic system using a compliance controller, the state data $S_{sub.t}$ includes robot position and velocity states, along with contact forces and torques. The reward data $R_{sub.t}$ is typically a single numerical value corresponding to the success of the assembly operation or lack thereof. For example, if the compliance controller/robot system fails to install the first part into the second part, the reward may be assigned a small negative value such as -1, and if the compliance controller/robot system successfully installs the first part into the second part, the reward may be assigned a large positive value such as 100.

(36) The agent **620** (the reinforcement learning module) may be modelled as a neural network. Upon receiving the state data $S_{sub.t}$ and the reward data $R_{sub.t}$, the agent **620** determines an action $A_{sub.t}$ which is believed to be a most effective input to the system (the environment **610**) to perform the assembly operation. In the case of a compliance controller/robot system, the action data $A_{sub.t}$ is an adjustment to the target position in six degrees of freedom (three orthogonal translations and three rotations). Upon receiving the action data $A_{sub.t}$, the compliance controller/robot system (the environment **610**) executes a next step of attempting the assembly operation, and provides as output a new value of the state data ($S_{sub.t+1}$) and a new value of the reward data ($R_{sub.t+1}$), and the agent **620** then determines a next action based on the new input. This continues until the assembly is successfully completed or ultimately fails.

(37) The agent **620** (the reinforcement learning module) uses the action data, the state data and the reward data to train the neural network to learn the most effective actions for a particular set of state data. This is done by constructing the reinforcement learning problem as having an objective to maximize the cumulative reward while considering randomness, using a formulation such as:

$$(38) \max \mathbb{E} \left[\sum_{t=0}^T \gamma^t r(s_t, a_t) \right] \quad (1)$$

Where the formulation attempts to maximize the expected cumulative value over an assembly operation (from time $t=0$ to $t=T$) of a reward r as a function of a state $s_{sub.t}$ and action $a_{sub.t}$, and where γ is a discount factor.

(39) In the techniques of the present disclosure, the reinforcement learning system is configured for training using an actor-critic formulation. In an actor-critic formulation, the actor is a stochastic feedback controller (as in FIG. 6 discussed above) which determines an action based on a state using a probability function, and the critic trains the actor to maximize the reward. The actor and critic are approximated by neural networks.

(40) The actor neural network determines an action a based on a policy π which is a function of the state s . That is:

$$(41) a \sim (\text{.Math. .Math. } s) = (\pi(s), \sigma(s)) \quad (2)$$

Where the actor policy π is represented by a Gaussian distribution N with a mean μ and a standard deviation σ . Training of the actor policy involves adjusting the parameters (the mean μ and the standard deviation σ) of the probability function to provide an action a which leads to the most effective results as determined by the reward. The actor does not directly use the reward value from the robot/controller, but the actor is trained by the critic which does use the reward.

(42) The critic is a function of state and action called a Q function, which is used in training of the actor. State, action, and reward data are required as input in order to train the critic. The critic neural network Q function is the approximation of cumulative reward (as in Equation (1)), and is trained to minimize a temporal difference error as follows:

$$(43) \min [Q(s, a) - (r + \gamma Q(s', a'))]^2 \quad (3)$$

Where $Q(s, a)$ is the critic function Q (of state s and action a), r is the reward for the current step, and the last term is the reward for the next step (s' , a') predicted by the Q function using a discount factor γ for cumulative reward calculation. The next action is determined by the actor policy; that is, $a' \sim \pi(\cdot | s')$.

(44) The actor neural network is then trained to maximize the estimated reward by the critic. To do this, the critic adjusts the parameters (μ , σ) of the actor policy π in order to maximize the value of the Q function:

$$\max Q(s, a) \quad (4)$$

Where the action a is determined by the policy π as defined above in Equation (2).

(45) Using the techniques discussed in the preceding paragraphs and in Equations (1)-(4), the actor and critic neural networks are trained by simply observing and discovering the response of the compliance controller/robot system to inputs, trying various input action strategies as dictated by the probabilistic nature of the actor and critic functions, and learning which action strategies lead to the most successful outcomes. At the very beginning, the neural networks are randomly initialized, and random actions are generated from the actor neural network to drive the robot. At each step in the assembly operation, the state s is measured and an action a is generated by the actor, then the action is evaluated and at the next step s' , the step reward r is obtained. The data in one step (s , a , s' , r) is recorded to a dataset called a replay buffer and then this process is repeated. There is a maximum allowed number of steps for one assembly operation; if the task succeeds or the maximum step number is reached, the assembly operation is finished. After performing one or several assembly operations, all of the data points (in the form of (s , a , s' , r)) are used to update the critic and actor neural networks.

(46) Returning to FIG. 5, and in accordance with the preceding discussion of an actor-critic reinforcement learning system, the actor block **520** and the critic block **530** operate as follows. On a line **522**, the actor block **520** provides an action, in the form of a target position correction, to the summing junction **440**. The action is determined based on the actor policy, as a function of states provided to the actor block **520** on a line **542**. The target position correction from the actor block **520** is also provided to the critic block **530** on a line **524**. This is because the critic function requires action data along with state and reward data for its training, as discussed earlier.

(47) The target position correction provided to the summing junction **440** has the effect of changing the target position used by the admittance control block **460**, effectively controlling the strategy used by the compliance controller/robot system in performing the assembly operation. The admittance control block **460** interacts with the robot **470** on a relatively high frequency control cycle (e.g., 125 Hz), and the robot **470** provides feedback to the reinforcement learning module **510** on a relatively low frequency control cycle (e.g., 10 Hz). In other words, the inner loop (the admittance control block **460** and the robot **470**) runs several or many steps before each step of the

outer loop (the reinforcement learning module **510** providing target position correction to the compliance controller/robot in the block **420**).

(48) The robot **470** provides the states (robot pose or position, robot velocity, and contact force/torque) to the actor block **520** on the line **542** as mentioned earlier. The states are also provided to the critic block **530** on a line **544**, as the critic requires state data. Reward data from the robot **470** (small negative reward if assembly step was unsuccessful, large positive reward if assembly step was successful) is provided to the critic block **530** on a line **554**. Thus, the critic block **530** receives all three of the inputs (action, state and reward) which it needs for training. The critic block **530** performs ongoing training of the actor block **520** by way of parameter adjustment (of the actor neural network) on a line **532**.

(49) Using the system illustrated in FIG. 5, training of the actor and critic neural networks (in the blocks **520** and **530**, respectively) is carried out using all data collected from the compliance controller/robot system during assembly operations in a learning mode. This continues until the performance of the actor block **520** reaches a satisfactory level.

(50) FIG. 7 is a block diagram illustration of a system **700** configured for robotic assembly skill learning, using a compliance controller and robot with a trained actor-only reinforcement learning module, according to an embodiment of the present disclosure. In FIG. 7, a reinforcement learning module **710** includes only the actor block **520** of FIG. 5. In this case, the actor block **520** (the actor neural network) has been trained as described above to optimize the success of the robotic assembly operation. Thus, the critic is no longer needed, and the parameters of the actor neural network are not further adjusted. FIG. 7 depicts the assembly skill learning system as it is used for actual productive work by the robot **100**. As before, the robot **470** provides state data on the line **542** to the actor block **520**, which determines an action based on the state data using the policy. The action (target position adjustment or correction) is provided to the summing junction **440** on the line **522**.

(51) The system illustrated in FIGS. 5 and 7 (compliance controller/robot system with reinforcement learning module) has been implemented and demonstrated to provide superior performance. For example, in a dual peg/hole installation, a compliance controller/robot with no supplemental control strategy had a success rate of about 43% over an assembly operation time of about 35 seconds (**350** attempted motion steps at 10 Hz). The same compliance controller/robot performing the same operation with an unguided supplemental searching strategy (as in FIG. 2) had a success rate of about 83% over an assembly operation time of about 19 seconds (**190** attempted motion steps). The same compliance controller/robot performing the same operation with the actor-critic reinforcement learning controller of the present disclosure had a success rate of 100% over an assembly operation time of about 12 seconds (**120** attempted motion steps).

(52) In one embodiment, the reinforcement learning module **510/710** is programmed into the robot controller **410** (along with the compliance control algorithm of the elements **430-460**). Other implementation embodiments are possible, where the reinforcement learning module **510/710** runs on a computer or cloud server separated from the robot controller **410**; for example, the reinforcement learning module **510/710** could be executed on a device which is configured for efficient neural network computation. Another alternative is to train the actor-critic system running on a separate computer, and then incorporate the trained actor-only reinforcement learning controller into the robot controller **410**. Any combination of these implementation embodiments may be used as suitable.

(53) FIG. 8 is a flowchart diagram **800** of a method for robotic assembly skill learning using actor-critic reinforcement learning with a compliance controller/robot system, according to an embodiment of the present disclosure. At box **802**, a compliance controller/robot system is provided, including an admittance control module controlling a robot performing an assembly task with force feedback used by the admittance control module to determine a next robot motion toward a target position and within a designated motion limit.

(54) At box **804**, an actor-critic reinforcement learning module in communication with the compliance controller/robot system is provided. The reinforcement learning module includes an actor neural network which receives state data feedback from the robot and determines an action in the form of a target position correction which is provided as input to the admittance control module. The reinforcement learning module also includes a critic neural network which receives state data feedback and reward data feedback from the robot, along with the action from the actor neural network. The critic neural network uses the action, the state data and the reward data for training to optimize the reward based on the state and action data. The critic neural network also adjusts parameters of the actor neural network to train the actor to provide actions which optimize the reward.

(55) At box **806**, the compliance controller/robot system and the reinforcement learning module are operated in a learning mode, where the robot system performs assembly tasks and the reinforcement learning module trains the actor and critic neural networks. At decision diamond **808**, it is determined whether the actor neural network has been sufficiently trained to reliably provide successful guidance of the robotic assembly operation. If not, learning mode continues.

(56) At box **810**, after the actor neural network demonstrates successful capability, the compliance controller/robot system and an actor-only version of the reinforcement learning module are operated in a production mode, where the robot system performs assembly tasks while being guided by target position corrections from the actor neural network. In the production mode at the box **808**, the critic neural network is no longer used, and the actor-only reinforcement learning module is used to provide feedback control to the compliance controller/robot system—with the actor neural network operating in inference mode.

(57) The actor-critic reinforcement learning technique described above provides significantly improved performance in robotic assembly operations using a compliance controller. The actor-critic reinforcement learning technique offers many advantages: it is fast, as it adjusts motion optimally based on force feedback; it is flexible, with no need to design or tune system parameters; it is easy to apply, as the reinforcement learning system learns optimal behavior by self-discovery using the real robotic system, and no model or human tuning are required; and it is a general approach which may be applied to simple or complex tasks.

(58) Another type of assembly skill learning system uses human demonstration of an operation to pre-train a reinforcement learning system, which then performs ongoing training in a self-learning mode with co-training provided by human override as needed to teach optimal behavior. This system is discussed below.

(59) FIG. **9** is a conceptual illustration of a system **900** for robot assembly skill learning combining human demonstration and reinforcement learning-based discovery, according to an embodiment of the present disclosure. A robot such as the robot **100** with a compliance controller is configured to perform an assembly operation, as discussed earlier.

(60) In a first step of the process (see circled number **1**), a human operator **910** demonstrates the assembly operation in cooperation with the robot **100**. One technique for demonstrating the operation involves putting the robot **100** in a teach mode, where the human **910** either manually grasps the robot gripper and workpiece and moves the workpiece into the installed position in the second workpiece (while the robot and controller monitor robot and force states), or the human **910** uses a teach pendant to provide commands to the robot **100** to complete the workpiece installation. Another technique for demonstrating the operation is teleoperation. In one form of teleoperation, the human **910** manipulates a duplicate copy of the workpiece which the robot **100** is grasping, and the human **910** moves the duplicate workpiece (which is instrumented and provides motion commands to the robot **100**) while watching the robot **100**, using the visual feedback from the robotic assembly operation and the human's own tactile feel to guide the successful completion of the assembly operation by the robot **100**. In another form of teleoperation, the human **910** uses a joystick-type input device to provide motion instructions (translations and rotations) to the robot

100. These or other human demonstration techniques may be used.

(61) The human demonstrator **910** preferably demonstrates the assembly operation several times, so that several complete sets of state and action data, each leading to successful installation, may be collected. The demonstration data (robot motion and force states, and actions, as discussed at length earlier) is collected in a database **920**.

(62) In a second step, the demonstration data from the database **920** is provided to a reinforcement learning system **930**. The reinforcement learning system **930** includes an agent (a neural network) which learns what actions are effective in correlation to a set of robot and force states, based on reward data. The demonstration data from the database **920** is provided to the reinforcement learning system **930** for pre-training. In this pre-training mode (the second step of the overall process, as indicated by the circled numeral **2**), there is no environment interacting with the agent as discussed earlier in connection with FIG. **6**. Instead, in the pre-training mode, the agent neural network in the reinforcement learning system **930** is trained purely based on the demonstration data from step one. Thus, the environment block and the feedback lines are all shown as dashed (not used) in the pre-training step.

(63) In a third step of the process, the pre-trained reinforcement learning system **930** and a compliance controller/robot system **940** are placed in an online production mode where self-learning occurs. The compliance controller/robot system **940** may use the same robot **100** as was used for human demonstration, or a separate instance of the same robot configuration. In the online self-learning mode, the compliance controller/robot system **940** repeatedly performs the prescribed installation (assembly operation) in a production mode, with additional control provided by the reinforcement learning system **930**. An example of the production (or “online”) mode of operation is where the second part is placed into a fixture (possibly by a second robot) or the second part arrives on a conveyor, the robot **100** picks up the first part and installs it into the second part, and the assembled unit moves on for another assembly step or for packaging, and then the process repeats with the next two parts, etc.

(64) As the reinforcement learning system **930** and the compliance controller/robot system **940** perform assembly operations in the third step, more cycles of learning data accumulate and are stored in a database **922**. The database **922** initially includes the human demonstration data from the database **920**, and data from the ongoing operation in the third step is added to the database **922**. The data includes the action, state and reward data needed for training the reinforcement learning system **930**, as discussed earlier.

(65) If the assembly operation is not particularly difficult, the reinforcement learning system **930** and the compliance controller/robot system **940** may run indefinitely in the online self-learning mode, with a very high success rate. This would be the case when the installation of the first part into the second part has a fairly loose tolerance, or the parts include geometric features which mechanically guide one part into the other, for example. However, in tight-tolerance assembly operations, some attempted installations may be unsuccessful, and failure data in the database **922** may start to overwhelm the reinforcement learning system **930**. That is, when positive reward data is sparse, the neural network in the agent of the reinforcement learning system **930** cannot properly correlate effective actions to given states.

(66) Because of the situation described above, in the techniques of the present disclosure, a fourth step is added for difficult assembly operations. The fourth step is a co-training mode where human correction is provided during the online self-learning mode. The human correction (co-training) phase includes monitoring the success rate of the assembly operations in the online self-learning mode. If the success rate drops below a predefined threshold, or attempted assembly operations exhibit searching behavior which is clearly off-base, then a human operator **950** steps in and interacts with the compliance controller/robot system **940** to override the reinforcement learning system **930**. The preferred mode of interaction between the human operator **950** and the compliance controller/robot system **940** is teleoperation, which was discussed above.

(67) By using the human intervention/correction step (co-training) described above, new successful learning cycles are added to the database **922**, such that the high reward values provide beneficial update training of the agent neural network in the reinforcement learning system **930** to identify effective actions for given states. The human intervention/correction step may be performed for a period of time, with the human operator **950** monitoring and intervening as necessary to ensure that each attempted assembly operation is successful. After this period of co-training, it would be expected that the reinforcement learning system **930** and the compliance controller/robot system **940** resume autonomous operation in the online self-learning production mode.

(68) FIG. **10** is a block diagram illustration of a system **1000** configured for robotic assembly skill learning, using a compliance controller with a reinforcement learning module, in a co-training mode for human correction during ongoing self-learning, according to an embodiment of the present disclosure. The usage of the system **1000** depicted in FIG. **10** for the co-training mode (step **4**) discussed above with respect to FIG. **9** is described below.

(69) At the top of FIG. **10** is the block **420** depicting the compliance controller interaction with the robotic assembly operation, as illustrated in FIG. **4** and discussed earlier. In the technique of the present disclosure, a reinforcement learning module **1010** is coupled to the compliance controller/robot system in the block **420**. The purpose of the reinforcement learning module **1010** is to learn an effective high level motion strategy for the assembly operation and use this strategy to guide the compliance controller/robot system. The reinforcement learning module **1010** corresponds with the reinforcement learning system **930** of FIG. **9**. For the purposes of this discussion, the database **920** and the agent neural network which learns the most effective assembly behavior are considered to be part of the reinforcement learning module **1010**, as shown in block **1020** of FIG. **10**.

(70) In the same manner as discussed with respect to FIG. **5**, the robot **470** provides state feedback (robot position and velocity, and contact force/torque between the parts) on a line **1042** to the block **1020**, and reward data (large positive reward for successful assembly; small negative reward for unsuccessful attempt) on a line **1044**. The agent block **1020** stores the state data and reward data in the database, and uses the data in training of the agent neural network in the reinforcement learning module **1010**.

(71) FIG. **10** depicts the system **1000** as configured for human correction/intervention (co-training), while FIG. **11** (discussed below) shows the same system **1000** as configured for self-learning without human co-training. When configured for human co-training input as in FIG. **10**, a human demonstrator **1050** uses a technique such as teleoperation to guide the assembly operation. A “switch” **1032** is positioned to receive motion data from the human demonstration (e.g., teleoperation) provided on a line **1052**.

(72) The motion data from the human demonstration is provided to the compliance controller/robot system in the block **420** as target position correction input on line **1062** to the summing junction **440**. During human co-training in the online self-learning phase, target position correction is provided both from human demonstration (on the line **1062**) and from the reinforcement learning module **1010** (from the agent **1020** on line **1022**), where the human demonstration data is preferably weighted higher than the reinforcement learning data. This arrangement enables the human demonstration input to guide the robot during the assembly operation.

(73) The motion data from the human demonstration may also be provided as action data on line **1064** to the agent **1020** in the reinforcement learning module **1010**. This connection provides all of the action data which is needed by the agent **1020** in order to learn the correlation between states and actions which lead to successful assembly operations.

(74) Thus, the configuration of FIG. **10** enables two things to happen; first, the human demonstration controls the actual motion of the compliance controller/robot system **420** during co-training; second, action data from the human demonstration is provided to the agent block **1020**, which along with the state data and reward data, allows the agent neural network to learn the

correlation between states and effective actions.

(75) The system of FIG. **10** could also be used in a simplified form for the human demonstration phase. This may be done because the connections exist from the human demonstrator **1050** to provide input to the compliance controller/robot system **420**, and from the robot **470** to provide state and reward feedback to the agent **1020** (which includes the database). If the system **1000** is used for the human demonstration phase (first step of FIG. **9**), only the human demonstration input is provided to the compliance controller in the block **420**, so the human demonstration exclusively controls the robot motion. Furthermore, the state, reward feedback and human demonstrated action are collected in the database **920** (discussed above), but the agent in the block **1020** does not provide an action to the summing junction **440**. After collecting the human demonstration data, offline pre-training of the agent in the block **1020** could then proceed as discussed earlier, without any interaction with the compliance controller/robot system in the block **420**.

(76) FIG. **11** is a block diagram illustration of the system **1000** configured for robotic assembly skill learning, using a compliance controller with a reinforcement learning module, in an online self-learning mode, according to an embodiment of the present disclosure. The system **1000** of FIG. **11** is the same as the system **1000** of FIG. **10** except in a different configuration.

(77) FIG. **11** depicts the system **1000** as configured for autonomous learning without human co-training input. This is the online self-learning performed in the third step of FIG. **9**. When configured for autonomous operation with no human demonstration input as in FIG. **11**, the human demonstrator **1050** is not needed, the switch **1032** is positioned so as not to receive motion data on the line **1052**, and only the action data on the line **1022** now flows to the summing junction **440**. No action data is provided to the summing junction **440** on the line **1062**, and no action data is provided back to the agent in the block **1020** on the line **1064**.

(78) Thus, the configuration of FIG. **11** enables two things to happen for autonomous operation of the reinforcement learning module **1010** with the compliance controller/robot system in the block **420**; first, the agent neural network in the block **1020** controls the actual motion of the compliance controller/robot system, in the online self-learning phase; second, action data is inherently known by the agent block **1020**, which along with the state data and reward data, allows the agent neural network to continuously learn the correlation between states and effective actions.

(79) The system **1000** in the configurations of FIGS. **10** and **11** therefore provide all of the capability needed for human demonstration, offline pre-training, online self-learning and human correction/intervention (co-training) as illustrated in the four steps of FIG. **9**. During online productive operations of the robot, the system **1000** can readily be switched between the configurations of FIGS. **10** and **11**—simply by opening or closing the switch **1032**—to allow human correction/intervention when needed, and return to autonomous operation of the reinforcement learning module **1010** when appropriate.

(80) The following discussion provides further explanation of the techniques used by the reinforcement learning module **1010** for learning and applying a policy to effectively control the compliance controller/robot system.

(81) The general principles of reinforcement learning were discussed above with respect to FIG. **6**. In typical reinforcement learning based approaches, low learning efficiency is a major drawback. Though reinforcement learning is well known for its capability to learn robust policy, the learning process requires a lot of interaction data if no a priori information is provided. When reward data is sparse, and/or for tasks in which the full state space is difficult to discover (e.g., precise insertion task, where there is very low chance to sample actions leading to successful insertion), an unacceptably high amount of interaction data is required to learn a good control policy. In an imitation learning based approach, less interaction data is required. However imitation learning results may be limited to a specific task. Also, the learning result is less robust to task variations since imitation learning does not generalize well to unseen situations.

(82) The techniques of the present disclosure were developed to overcome the shortcomings

described above. The first part of the presently disclosed technique uses offline pre-training of the reinforcement learning system. The reason to introduce offline reinforcement learning is that it is much faster than standard reinforcement learning. Standard reinforcement learning approaches require continuous interaction with the “environment” (the compliance controller/robot system) for data collection along with training. Typically the interactive data collection process is much slower than training. It is possible for offline reinforcement learning to perform training on a given dataset (i.e., the demonstration data) without further interactive data collection, and thus offline reinforcement learning takes less time for pre-training.

(83) The working principle of offline reinforcement learning is similar to standard reinforcement learning approaches. However, if standard reinforcement learning approaches are used to train a control policy on a given dataset without further interactive data collection, the training process becomes unstable. During the training process, overestimation of the Q function is inevitable, especially in unseen state-action space. This overestimation leads to blind optimism of unseen action space, and further leads to positive feedback of the overestimation of the Q function in unseen state-action space, which finally causes divergence of the entire training process. In the presently disclosed offline reinforcement learning pre-training, this issue is addressed by introducing an extra constraint to the learning formulation (i.e., either to the critic (Q function) or the actor in an actor-critic reinforcement learning system) to guarantee that the updated policy does not deviate too much from the demonstration dataset.

(84) One example of a constraint on the actor of an actor-critic reinforcement learning system is given as:

(85)

$$\max_{a \sim (\cdot | \text{Math. } S)} [Q(s, a) - \lambda D_{\text{KL}}(\pi(\cdot | s) \parallel \pi_{\text{demo}}(\cdot | s))] \quad (5)$$

(86) In the optimization problem above, the objective function includes a loss function computed by subtracting a divergence term from the reward computed by the Q function. In the presently disclosed technique, the loss function includes a Kullback-Leibler divergence calculation (D_{KL}) which is a measure of how one probability distribution [the control policy of the actor, $\pi(\cdot | s)$] is different from a second, reference probability distribution [the training dataset, $\pi_{\text{demo}}(\cdot | s)$]. In Equation (5), λ is a weighting constant. By subtracting the divergence term from the Q function reward term, the objective function penalizes behavior of the actor control policy π which deviates from the training dataset. With this training constraint, overestimation can be avoided and the training process becomes stable on a given dataset.

(87) When the pre-training of the control policy is done using the demonstration data as described above, the training process moves to the self-learning stage (step 3 of FIG. 9). In the self-learning stage, the robot further interacts with the assembly task for evaluation of control performance and data collection. The collected interaction data is further utilized to fine-tune the control policy in the agent neural network for performance improvement. In this stage, the human operator (950, 1050) is able to further assist the learning process by providing correction or guidance (co-training), as illustrated in FIGS. 9 and 10. For a task with process uncertainty, it is not possible for the limited human demonstration (step 1) to cover all task variations. Thus, the pre-trained controller may perform poorly if a task variation is not covered in the demonstration dataset. In this situation, the standard reinforcement learning training progress relies on random discovery, and lots of random trials are needed. In the techniques of the present disclosure, a self-learning mode (co-training) is included. In co-training mode, the human operator is able to override the robot command if a poor action is generated by the learning controller during the self-learning stage. With this guidance mechanism, proper state-action pairs can be recorded in the training dataset with less interaction and the learning controller can learn to improve its performance within a short time.

(88) Using the techniques described above, the systems depicted in FIGS. 9-11 have been

demonstrated to produce dramatically improved performance relative to standard or naïve reinforcement learning approaches and also relative to traditional imitation learning approaches. Using the techniques of the present disclosure, robotic assembly operations on a variety of workpiece shapes have been demonstrated to have higher success rates and shorter cycle times than the same operations performed using prior art methods.

(89) FIG. 12 is a flowchart diagram 1200 of a method for robotic assembly skill learning using a reinforcement learning controller with a compliance controller/robot system, including offline pre-training using human demonstration data and online self-learning with human co-training, according to an embodiment of the present disclosure. At box 1202, a compliance controller/robot system is provided, including an admittance control module controlling a robot performing an assembly task with force feedback used by the admittance control module to determine a next robot motion toward a target position within a designated motion limit.

(90) At box 1204, a reinforcement learning module in communication with the compliance controller/robot system is provided. The reinforcement learning module includes at least one neural network which receives state data feedback from the robot and determines an action in the form of a target position correction which is provided as input to the admittance control module. The reinforcement learning module also receives reward data feedback from the robot. The neural network in the reinforcement learning module (or the two neural networks in an actor-critic system) uses the state data and the reward data for training to optimize the reward based on the state data and corresponding action data.

(91) At box 1206, human demonstration of the prescribed assembly operation is performed and pre-training of the reinforcement learning module is completed using the data from human demonstration. The box 1206 represents the first and second steps (the top half) of FIG. 9. As discussed earlier, several repetitions of the human demonstration are preferably performed, with the action, state and reward data captured in the demonstration database 920. The demonstration data in the database 920 is then used to perform initial pre-training of the reinforcement learning agent, without any interaction with the compliance controller/robot system. The pre-training includes using an optimization objective function which constrains the neural network in the reinforcement learning controller to behavior as defined in the demonstration data.

(92) At box 1208, the compliance controller/robot system with the reinforcement learning module is placed online in self-learning mode. This is the third step of FIG. 9, where the robot with compliance controller repeatedly performs the assembly operation with additional feedback control by the reinforcement learning system 930. In the self-learning mode at the box 1208, the agent neural network in the reinforcement learning module continues to learn as data (action, state, reward) is collected in the database 922 (with initial data from the database 920). This self-learning mode was also depicted in FIG. 11, using the reinforcement learning module 1010 providing feedback control to the compliance controller/robot system in the block 420.

(93) At decision diamond 1210, it is determined whether the performance of the compliance controller/robot system with the reinforcement learning module is acceptable. The performance is evaluated by a series of assembly operations, and key performance index is measured. The performance index can be designed to include combinations of operation success rate, operation cycle time, and other factors determined by the operator. If performance is acceptable, online operations continue in self-learning mode at the box 1208.

(94) If performance is not acceptable at the decision diamond 1210, then online operations switch to co-training mode at box 1212. Co-training—where the human operator uses teleoperation to override the input to the compliance controller/robot system—was depicted in the fourth step of FIG. 9 and also in FIG. 10. During co-training, the reinforcement learning module receives the necessary action, state and reward data to continue learning, and the reinforcement learning module may also contribute a small amount of input to the compliance controller/robot system. The co-training by the human operator may continue for several assembly operations and/or for a certain

period of time, at which point the performance of the system in self-learning mode is again evaluated at the decision diamond **1210**. As discussed earlier, the co-training phase at the box **1212** produces high-reward training data from successful assembly operations, thereby providing additional training of the neural network in the reinforcement learning module to build the correlation between states and effective actions.

(95) The system of FIGS. **9-11**, combining imitation learning (by human demonstration) and reinforcement learning (self-learning with human correction, or co-training), provides many advantages in robot skill learning. The system is flexible—being generally applicable to many different part types and shapes without modification. The system is also easy and intuitive—requiring no programming or delicate fine-tuning of parameters. The system can be applied to relatively simple assembly tasks where little or no human intervention will be needed for ongoing self-learning operations, and/or to tight-tolerance difficult assembly tasks where the use of human correction/intervention as needed ensures the necessary high success rates.

(96) Throughout the preceding discussion, various computers and controllers are described and implied. It is to be understood that the software applications and modules of these computers and controllers are executed on one or more computing devices having a processor and a memory module. In particular, this includes a processor in the robot controller **410** which controls the robot **100** performing the robotic assembly task as shown in FIG. **4**, and the robot controllers and optional other computers which execute the functions of the compliance controller block **420** and the various reinforcement learning modules depicted in FIGS. **4-11**.

(97) The foregoing discussion discloses and describes merely exemplary embodiments of the present disclosure. One skilled in the art will readily recognize from such discussion and from the accompanying drawings and claims that various changes, modifications and variations can be made therein without departing from the spirit and scope of the disclosure as defined in the following claims.

Claims

1. A method for robotic assembly skill learning, said method comprising: providing a robot controlled by a compliance controller, the robot being configured to perform an assembly operation where the robot moves a first part into an assembled position with a second part; providing a reinforcement learning controller in communication with the compliance controller and the robot, said reinforcement learning controller having a neural network which defines a policy for determining an action in response to state data received as feedback from the robot; pre-training the reinforcement learning controller, including using a human demonstration dataset to train the neural network in the reinforcement learning controller to cause the policy to maximize reward data received as feedback from the robot; operating the robot to perform the assembly operation with the reinforcement learning controller in a self-learning mode, where the reinforcement learning controller provides the action as input to the compliance controller, and the state data and the reward data received as feedback from the robot are used to continuously train the neural network in the reinforcement learning controller; and periodically operating the robot to perform the assembly operation with the reinforcement learning controller in a co-training mode, where a human demonstrator provides supplemental input to the compliance controller to control the robot to perform the assembly operation.
2. The method according to claim 1 wherein the action is a target position adjustment including three translations and three rotations of the first part relative to the second part which is provided to the compliance controller, and where the target position adjustment is combined with a nominal target position, bounded by a predefined motion limit, and used as the input to the compliance controller.
3. The method according to claim 1 wherein the state data includes robot positions and velocities

each including three translational and three rotational components, and part-to-part contact forces and torques each in three directions.

4. The method according to claim 1 wherein the reward data includes a positive value when the assembly operation is successfully completed and a negative value when the assembly operation is not successfully completed, and the positive value has a magnitude greater than the negative value.
5. The method according to claim 1 wherein the policy defined by the neural network in the reinforcement learning controller is a statistical distribution of actions relative to states, where the statistical distribution is defined by parameters including a mean and a standard deviation.
6. The method according to claim 1 wherein the human demonstration dataset includes action, state and reward data captured for multiple human demonstrations of the assembly operation, and pre-training the reinforcement learning controller includes using the human demonstration dataset for training with no interaction between the reinforcement learning controller and the robot.
7. The method according to claim 1 wherein pre-training the reinforcement learning controller includes an offline reinforcement learning technique using a Kullback-Leibler divergence calculation in a loss function which penalizes deviation of the policy learned by the neural network from a policy based on the human demonstration dataset.
8. The method according to claim 1 wherein the co-training mode includes using human demonstration by teleoperation and is invoked when a robot success performance metric drops below a predefined threshold, and action data from the teleoperation along with the state data and the reward data received as feedback from the robot in the co-training mode are used to further train the neural network in the reinforcement learning controller.
9. The method according to claim 1 wherein the reinforcement learning controller has an actor module including the neural network and a critic module including a second neural network, where the actor module provides the action as input to the compliance controller and the critic module updates parameters of the policy of the actor module based on a critic function.
10. The method according to claim 9 wherein training of the actor module by the critic module includes using an optimization computation which determines desired actions which maximize the reward calculated by the critic function, and adjusting the parameters of the policy in the actor module based on the desired actions.
11. The method according to claim 9 wherein the critic function computes a reward for a current state and a current action as equal to a current actual reward plus a term including a weighted expected reward for a future state and a future action.
12. The method according to claim 1 wherein the compliance controller controls the robot in a feedback loop having a first cycle frequency, the reinforcement learning controller controls the compliance controller in a feedback loop having a second cycle frequency, and the first cycle frequency is higher than the second cycle frequency.
13. The method according to claim 1 wherein the compliance controller and the reinforcement learning controller are both executed on a robot controller device which provides joint motion commands to the robot and receives the state data as feedback from the robot.
14. A robotic assembly skill learning system, said system comprising: a robot controlled by a compliance controller, the robot being configured to perform an assembly operation where the robot moves a first part into an assembled position with a second part; and a reinforcement learning controller in communication with the compliance controller and the robot, said reinforcement learning controller having a neural network which defines a policy for determining an action in response to state data received as feedback from the robot, where the reinforcement learning controller is pre-trained using a human demonstration dataset to train the neural network in the reinforcement learning controller to cause the policy to maximize reward data received as feedback from the robot, and after pre-training, the robot performs the assembly operation with the reinforcement learning controller in a self-learning mode, where the reinforcement learning controller provides the action as input to the compliance controller, and the state data and the

reward data received as feedback from the robot are used to continuously train the neural network in the reinforcement learning controller, and where the robot periodically performs the assembly operation with the reinforcement learning controller in a co-training mode, where a human demonstrator provides supplemental input to the compliance controller to control the robot to perform the assembly operation.

15. The system according to claim 14 wherein the action is a target position adjustment including three translations and three rotations of the first part relative to the second part which is provided to the compliance controller, and where the target position adjustment is combined with a nominal target position, bounded by a predefined motion limit, and used as the input to the compliance controller.

16. The system according to claim 14 wherein the state data includes robot positions and velocities each including three translational and three rotational components, and part-to-part contact forces and torques each in three directions, and the reward data includes a positive value when the assembly operation is successfully completed and a negative value when the assembly operation is not successfully completed.

17. The system according to claim 14 wherein the human demonstration dataset includes action, state and reward data captured for multiple human demonstrations of the assembly operation, and pre-training the reinforcement learning controller includes using the human demonstration dataset for training with no interaction between the reinforcement learning controller and the robot.

18. The system according to claim 14 wherein pre-training the reinforcement learning controller includes an offline reinforcement learning technique using a Kullback-Leibler divergence calculation in a loss function which penalizes deviation of the policy learned by the neural network from a policy based on the human demonstration dataset.

19. The system according to claim 14 wherein the co-training mode includes using human demonstration by teleoperation and is invoked when a robot success performance metric drops below a predefined threshold, and action data from the teleoperation along with the state data and the reward data received as feedback from the robot in the co-training mode are used to further train the neural network in the reinforcement learning controller.

20. The system according to claim 14 wherein the reinforcement learning controller has an actor module including the neural network and a critic module including a second neural network, where the actor module provides the action as input to the compliance controller and the critic module updates parameters of the policy of the actor module based on a critic function.

21. The system according to claim 20 wherein training of the actor module by the critic module includes using an optimization computation which determines desired actions which maximize the reward calculated by the critic function, and adjusting the parameters of the policy in the actor module based on the desired actions.

22. The system according to claim 20 wherein the critic function computes a reward for a current state and a current action as equal to a current actual reward plus a term including a weighted expected reward for a future state and a future action.

23. The system according to claim 14 wherein the compliance controller controls the robot in a feedback loop having a first cycle frequency, the reinforcement learning controller controls the compliance controller in a feedback loop having a second cycle frequency, and the first cycle frequency is higher than the second cycle frequency.

24. The system according to claim 14 wherein the compliance controller and the reinforcement learning controller are both executed on a robot controller device which provides joint motion commands to the robot and receives the state data as feedback from the robot.
